Solution to Challenge 9

Synchronization of Chaotic Orbits

(Alligood, Sauer & Yorke, *Chaos: An Introduction to Dynamical Systems*, Ch. 9)

Overview. Two copies of a system are coupled so that each is pulled toward the other. The *difference* $u = v_1 - v_2$ obeys its own equation, and the coupling adds a uniform damping $-2c\,u$ to it. If the damping $2c$ beats the rate λ at which the uncoupled system stretches differences, the difference dies out: the two copies synchronize. Steps 1–2 do this for $\dot{x} = ax$, Steps 3–5 for linear systems, Steps 6–7 for nonlinear systems (the Synchronization Theorem), and Step 8 for one-way coupling of two Lorenz systems.

Notation: $|\cdot|$ is the Euclidean norm, $\|\cdot\|$ the induced matrix norm (largest singular value).

Step 1: The scalar coupled system

System (9.27) is $\dot{x} = ax + c(y - x)$, $\dot{y} = ay + c(x - y)$, i.e.

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = A\begin{pmatrix}x\\ y\end{pmatrix}, \qquad A = \begin{pmatrix}a-c & c\\ c & a-c\end{pmatrix}.$$

Eigenvalues and eigenvectors.

$$A\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}a-c+c\\ c+a-c\end{pmatrix} = a\begin{pmatrix}1\\1\end{pmatrix}, \qquad A\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}a-2c\\ -(a-2c)\end{pmatrix} = (a-2c)\begin{pmatrix}1\\-1\end{pmatrix}.$$

So the eigenvalues are $\lambda_1 = a$ (eigenvector $(1,1)^T$, the "synchronized" direction $x = y$) and $\lambda_2 = a - 2c$ (eigenvector $(1,-1)^T$, the "difference" direction).

Diagonalizing. Let $S = \begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$, so $S^{-1} = \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$, and define (calling the new variable w to avoid confusion with the difference u used later)

$$w = \begin{pmatrix}w_1\\ w_2\end{pmatrix} = S^{-1}\begin{pmatrix}x\\ y\end{pmatrix} = \begin{pmatrix}(x+y)/2\\ (x-y)/2\end{pmatrix}.$$

Then $\dot{w} = S^{-1}AS\,w = \begin{pmatrix}a & 0\\ 0 & a-2c\end{pmatrix} w$, i.e. the equations decouple:

$$\dot{w}_1 = aw_1, \qquad \dot{w}_2 = (a-2c)w_2 \quad\Longrightarrow\quad w_1(t) = w_1(0)e^{at}, \quad w_2(t) = w_2(0)e^{(a-2c)t}.$$

Step 2: The solution and the difference

Since $x = w_1 + w_2$, $y = w_1 - w_2$, $w_1(0) = \frac{x_0+y_0}{2}$, $w_2(0) = \frac{x_0-y_0}{2}$:

$$\boxed{x(t) = \tfrac{x_0+y_0}{2}e^{at} + \tfrac{x_0-y_0}{2}e^{(a-2c)t}, \qquad y(t) = \tfrac{x_0+y_0}{2}e^{at} - \tfrac{x_0-y_0}{2}e^{(a-2c)t}.}$$

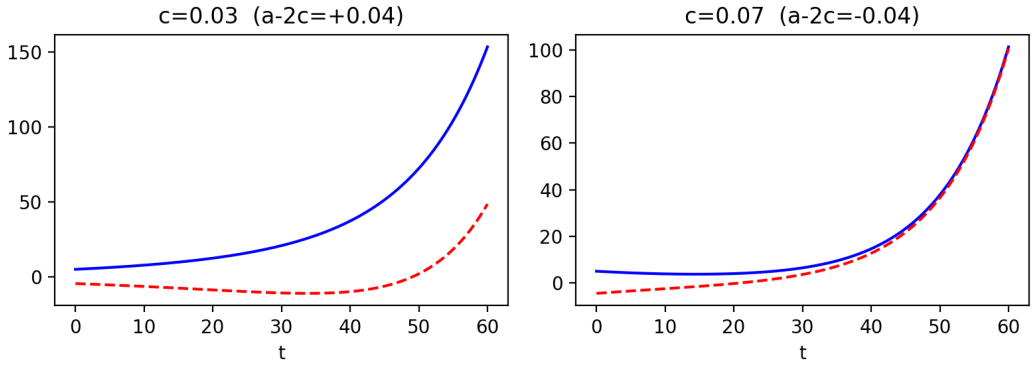
Subtracting,

$$x(t) - y(t) = (x_0 - y_0)\, e^{(a-2c)t} \quad \Longrightarrow \quad \boxed{|x(t) - y(t)| = |x_0 - y_0|\, e^{(a-2c)t}.}$$

Interpretation. The common part $\frac{x+y}{2}$ grows like e^{at} (each system is unstable), but the difference changes at rate $a - 2c$:

$$\begin{aligned} c > \tfrac{a}{2} &\;\Rightarrow\; |x - y| \to 0 \text{ (global synchronization)}, \\ c = \tfrac{a}{2} &\;\Rightarrow\; |x - y| \text{ constant}, \\ c < \tfrac{a}{2} &\;\Rightarrow\; |x - y| \to \infty. \end{aligned}$$

This explains Figure 9.13: with $a = 0.1$, the weak coupling $c = 0.03$ gives $a - 2c = +0.04 > 0$ (the difference grows), while $c = 0.07$ gives $a - 2c = -0.04 < 0$ (it shrinks like $9.5\, e^{-0.04t}$ for initial values 5 and -4.5). The switching value is $c = a/2 = 0.05$.



Numerical solutions of (9.27) with $a = 0.1$, $(x_0, y_0) = (5, -4.5)$: $c = 0.03$ (left) and $c = 0.07$ (right).

Step 3: Coupled linear systems: the eigenvalues

Let A be $n \times n$ and

$$M = \begin{pmatrix} A - cI & cI \\ cI & A - cI \end{pmatrix} \qquad (2n \times 2n),$$

the matrix of (9.30).

Eigenvectors. Suppose $Av = \lambda v$. Then

$$M \begin{pmatrix} v \\ v \end{pmatrix} = \begin{pmatrix} (A - cI)v + cv \\ cv + (A - cI)v \end{pmatrix} = \begin{pmatrix} Av \\ Av \end{pmatrix} = \lambda \begin{pmatrix} v \\ v \end{pmatrix},$$

$$M \begin{pmatrix} v \\ -v \end{pmatrix} = \begin{pmatrix} (A - cI)v - cv \\ cv - (A - cI)v \end{pmatrix} = \begin{pmatrix} (A - 2cI)v \\ -(A - 2cI)v \end{pmatrix} = (\lambda - 2c) \begin{pmatrix} v \\ -v \end{pmatrix}.$$

So $(v, v)^T$ and $(v, -v)^T$ are eigenvectors of M with eigenvalues λ and $\lambda - 2c$. If A has n independent eigenvectors $v_1, \ldots, v_n$, the $2n$ vectors $(v_i, v_i)^T, (v_i, -v_i)^T$ are independent (they are the columns of $Q\,\mathrm{diag}(V, V)$ with $Q = \begin{pmatrix} I & I \\ I & -I \end{pmatrix}$ and $V = [v_1 \cdots v_n]$, both invertible), so they give all of the eigenvalues $\lambda_1, \ldots, \lambda_n, \lambda_1 - 2c, \ldots, \lambda_n - 2c$.

The general case (any A, with multiplicities). Let $Q = \begin{pmatrix} I & I \\ I & -I \end{pmatrix}$, $Q^{-1} = \frac{1}{2}Q$. Then

$$MQ = \begin{pmatrix} A & A-2cI \\ A & -(A-2cI) \end{pmatrix} = Q \begin{pmatrix} A & 0 \\ 0 & A-2cI \end{pmatrix}, \qquad \text{so} \qquad M = Q \begin{pmatrix} A & 0 \\ 0 & A-2cI \end{pmatrix} Q^{-1}.$$

Similar matrices have the same characteristic polynomial, hence $\det(\mu I - M) = \det(\mu I - A)\,\det\big(\mu I - A + 2cI\big)$. Therefore the eigenvalues of M, counted with multiplicity, are exactly

$$\lambda_1, \dots, \lambda_n,\ \lambda_1 - 2c, \dots, \lambda_n - 2c. \qquad \blacksquare$$

(For $n = 1$ this is Step 1.)

Step 4: The difference equation

With $\dot v_1 = Av_1 + c(v_2 - v_1)$, $\dot v_2 = Av_2 + c(v_1 - v_2)$ and $u = v_1 - v_2$,

$$\dot u = A(v_1 - v_2) + c(v_2 - v_1) - c(v_1 - v_2) = Au - 2c\,u,$$

that is

$$\boxed{\dot u = (A - 2cI)u.} \tag{1}$$

(Likewise the sum $s = v_1 + v_2$ satisfies $\dot s = As$: the coupling terms cancel.)

Step 5: Global asymptotic stability, global synchronization

Claim. If every eigenvalue of A has real part $< 2c$, the origin of (1) is globally asymptotically stable.

Proof. The matrix $B = A - 2cI$ has eigenvalues $\lambda_i - 2c$, all with negative real part. Write $B = PJP^{-1}$ in Jordan form. Each entry of e^{Jt} is a constant times $\frac{t^k}{k!}e^{\mu t}$ with μ an eigenvalue of B, $\operatorname{Re}\mu < 0$. Since $t^k e^{\operatorname{Re}\mu\, t} \to 0$, every entry tends to 0, and so does $e^{Bt} = Pe^{Jt}P^{-1}$. More precisely, choosing $0 < \beta < \min_i\big(2c - \operatorname{Re}\lambda_i\big)$ there is a constant $C \geq 1$ with

$$\|e^{Bt}\| \leq Ce^{-\beta t} \qquad (t \geq 0).$$

The solution with $u(0) = u_0$ is $u(t) = e^{Bt}u_0$, so $|u(t)| \leq Ce^{-\beta t}|u_0|$.

- *Lyapunov stable:* given $\varepsilon > 0$ take $\delta = \varepsilon/C$; $|u_0| < \delta \Rightarrow |u(t)| < \varepsilon$ for all $t \geq 0$.
- *Globally attracting:* $|u(t)| \to 0$ for *every* $u_0 \in \mathbb{R}^n$.

This is global asymptotic stability. $\blacksquare$

Conclusion. For any initial values $v_1(0), v_2(0)$, $u(0) = v_1(0) - v_2(0)$ is arbitrary, so

$$\lim_{t\to\infty} |v_1(t) - v_2(t)| = 0 \qquad \text{whenever} \qquad c > \tfrac{1}{2}\max_i \operatorname{Re}\lambda_i.$$

This is global synchronization. (It does not say v_1, v_2 stay bounded: if some $\operatorname{Re}\lambda_i > 0$ they both blow up, but together; for $n = 1$ it recovers $c > a/2$.)

Step 6: The nonlinear case: $K(t) = J(t)e^{-2ct}$

Let $\dot{v} = f(v)$ (9.32) with flow F_t. Fix a trajectory $v(t) = F_t(v_0)$ and let

$$J(t) = DF_t(v_0), \qquad A(t) = Df\big(v(t)\big).$$

Differentiating $\frac{d}{dt}F_t(v_0) = f(F_t(v_0))$ with respect to v_0 gives the variational equation

$$J'(t) = A(t)J(t), \qquad J(0) = I \tag{2}$$

(eq. (9.15)).

Where K comes from. For the coupled pair (9.33) near the synchronized solution $v_1 = v_2 = v(t)$, perturbations (δ_1, δ_2) obey the variational equation (9.34): $\dot{\delta}_i = (A(t) - cI)\delta_i + c\,\delta_j$ $(j \neq i)$. Exactly as in Step 4, the difference $\delta = \delta_1 - \delta_2$ satisfies

$$\dot{\delta} = \big(A(t) - 2cI\big)\delta. \tag{3}$$

(In the nonlinear equation, $\dot{u} = f(v_1) - f(v_2) - 2cu$, and $f(v_1) - f(v_2) = A(t)u + O(|u|^2)$ for small u, which is where (3) comes from.) Let $K(t)$ be the Jacobian of the time-t map of (3), i.e. its fundamental matrix:

$$K'(t) = \big(A(t) - 2cI\big)K(t), \qquad K(0) = I.$$

Claim: $K(t) = J(t)e^{-2ct}$.

Proof. Let $L(t) = J(t)e^{-2ct}$. By the product rule and (2),

$$L'(t) = J'(t)e^{-2ct} - 2cJ(t)e^{-2ct} = A(t)J(t)e^{-2ct} - 2cJ(t)e^{-2ct} = \big(A(t) - 2cI\big)L(t),$$

and $L(0) = J(0) = I$. So L and K solve the same linear initial value problem; by uniqueness $L = K$. ∎

Step 7: The Synchronization Theorem

Theorem 9.7. Let λ_1 be the largest Lyapunov exponent of (9.32) (along a bounded orbit). If $c > \lambda_1/2$, then the coupled system (9.33) is locally synchronized: the synchronized state $u = v_1 - v_2 = 0$ is stable.

What the hypothesis means. By Definition 9.2, the Lyapunov exponents of the flow are those of the time-1 map F_1. For the orbit of v_0, the chain rule gives $D(F_1^n)(v_0) = DF_n(v_0) = J(n)$, and the ellipse into which a small sphere is carried after n steps has longest semi-axis $\|J(n)\|$. The largest exponent is therefore

$$\lambda_1 = \lim_{n\to\infty} \frac{1}{n} \ln\|J(n)\|.$$

Proof. **(a) At integer times.** By Step 6, $K(n) = J(n)e^{-2cn}$, so

$$\frac{1}{n}\ln\|K(n)\| = \frac{1}{n}\ln\|J(n)\| - 2c \;\longrightarrow\; \lambda_1 - 2c < 0.$$

(Indeed the same factor e^{-2cn} shifts *every* Lyapunov exponent: the exponents of the difference flow are $\lambda_i - 2c$.) Fix $\varepsilon > 0$ with $\beta := 2c - \lambda_1 - \varepsilon > 0$. There is C_1 with $\|J(n)\| \le C_1 e^{(\lambda_1+\varepsilon)n}$ for all $n \ge 0$ (the limit controls large n; finitely many small n are absorbed into C_1), hence

$$\|K(n)\| \le C_1 e^{-\beta n}.$$

(b) Between integer times. For $t \in [n, n+1]$ we have $F_t = F_{t-n} \circ F_n$, so $J(t) = DF_{t-n}\big(v(n)\big)J(n)$. The orbit is bounded, so $M := \sup\{\|DF_s(w)\| : 0 \le s \le 1,\ w \in \overline{\{v(t)\}}\} < \infty$ (continuity of $DF_s(w)$ on a compact set). Thus

$$\|K(t)\| = \|J(t)\|e^{-2ct} \le M\|J(n)\|e^{-2cn}e^{-2c(t-n)} \le MC_1e^{-\beta n} \le Ce^{-\beta t}, \qquad C = MC_1e^{\beta}.$$

(c) Stability. Every solution of the difference (variational) equation (3) is $\delta(t) = K(t)\delta(0)$, so

$$|\delta(t)| \le Ce^{-\beta t}|\delta(0)| \qquad (t \ge 0).$$

Hence $|\delta(t)| \le C|\delta(0)|$ for all t (Lyapunov stability: small differences stay small) and $|\delta(t)| \to 0$ exponentially (asymptotic stability). So the equilibrium $u = 0$ of the difference equation is stable, i.e. two copies started close together and coupled with $c > \lambda_1/2$ synchronize. ■

Remark. (1) For the linear system $\dot v = Av$, $J(t) = e^{At}$ and $\lambda_1 = \max \operatorname{Re} \lambda_i$, so the theorem reduces to Step 5. (2) With the Chua circuit, $\lambda_1 \approx 0.48$, the threshold is $c > 0.24$: the book's $c = 0.15$ does not synchronize while $c = 0.30$ does. (3) As in the book, the passage from the linearized difference equation to the full nonlinear one is the usual linearization step: the nonlinear remainder is $O(|u|^2)$ and cannot defeat exponential decay of the linear part for small u. Without coupling ($c = 0$) the same computation says nothing good: $\lambda_1 > 0$ means differences grow, i.e. sensitive dependence.

Step 8: One-way coupling of Lorenz systems

Sender (9.35): $\dot x_1 = -\sigma x_1 + \sigma y_1$, $\dot y_1 = -x_1z_1 + rx_1 - y_1$, $\dot z_1 = x_1y_1 - bz_1$. Receiver (9.36): $\dot y_2 = -x_1z_2 + rx_1 - y_2$, $\dot z_2 = x_1y_2 - bz_2$ (driven by the sender's signal $x_1(t)$). Let $u_y = y_1 - y_2$, $u_z = z_1 - z_2$.

The difference equations (exact, no linearization).

$$\dot u_y = (-x_1z_1 + rx_1 - y_1) - (-x_1z_2 + rx_1 - y_2) = -x_1(z_1 - z_2) - (y_1 - y_2) = -u_y - x_1u_z,$$
$$\dot u_z = (x_1y_1 - bz_1) - (x_1y_2 - bz_2) = x_1(y_1 - y_2) - b(z_1 - z_2) = x_1u_y - bu_z.$$

Thus

$$\dot u_y = -u_y - x_1(t)\,u_z, \qquad \dot u_z = x_1(t)\,u_y - b\,u_z, \tag{4}$$

a linear system whose coefficient $x_1(t)$ is the sender's chaotic signal (a bounded, hence continuous and globally defined, function of t); $u = 0$ is an equilibrium. Note that σ and r have disappeared.

The Lyapunov function. Let $E(u_y, u_z) = u_y^2 + u_z^2$. Then $E > 0$ for $u \ne 0$, $E(0) = 0$, $E \to \infty$ as $|u| \to \infty$, and

$$\begin{aligned}\dot E &= 2u_y\dot u_y + 2u_z\dot u_z = 2u_y(-u_y - x_1u_z) + 2u_z(x_1u_y - bu_z)\\ &= -2u_y^2 - 2x_1u_yu_z + 2x_1u_yu_z - 2bu_z^2 = -2u_y^2 - 2b\,u_z^2.\end{aligned}$$

The cross terms cancel *whatever* $x_1(t)$ *is*. Since $b > 0$, $\dot E < 0$ for every $u \ne 0$ and $\dot E = 0$ only at $u = 0$: E is a strict Lyapunov function, and this holds uniformly in t because the sign of $\dot E$ does not involve x_1.

Global synchronization. The system (4) is not autonomous (it contains $x_1(t)$), so we make the Lyapunov argument quantitative. With $\kappa = \min\{1, b\} > 0$, $\dot E = -2u_y^2 - 2bu_z^2 \le -2\kappa E$, hence

$$E(t) \le E(0)\,e^{-2\kappa t}, \qquad \text{i.e.} \qquad |u(t)| \le |u(0)|\,e^{-\kappa t} \quad (t \ge 0).$$

For *every* pair of initial conditions, (receiver arbitrary, sender anywhere),

$$y_2(t) - y_1(t) \to 0 \quad \text{and} \quad z_2(t) - z_1(t) \to 0 \quad \text{exponentially.} \qquad \blacksquare$$

(Stability, too: $|u(t)| \le |u(0)|$.) The condition $b > 0$ is what makes the z-direction contract; σ, r are arbitrary, in particular the chaotic values $\sigma = 10$, $r = 28$, $b = 8/3$.

What about x_2? The y_2- and z_2-equations contain only x_1, never x_2, so the pair (y_2, z_2) is what actually synchronizes. As printed in the book, $\dot{x}_2 = -\sigma x_1 + \sigma y_2$, with $u_x = x_1 - x_2$:

$$\dot{u}_x = \sigma(y_1 - y_2) = \sigma u_y, \qquad u_x(t) = u_x(0) + \sigma \int_0^t u_y \, ds \;\longrightarrow\; u_x(0) + \sigma \int_0^\infty u_y \, ds,$$

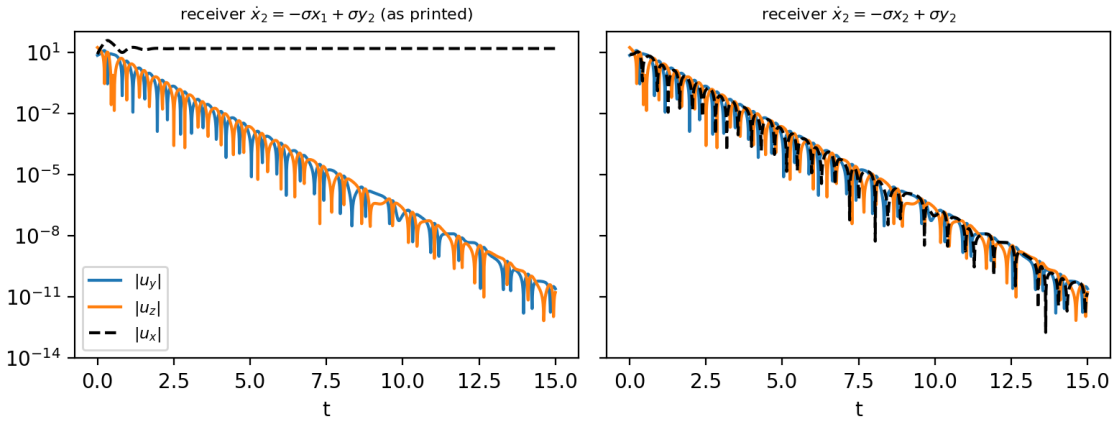
a finite limit (since $|u_y| \le |u(0)|e^{-\kappa t}$) which is generally *not* 0; so with this literal equation x_2 does not converge to x_1 for all initial data (a numerical run with $(x_1, y_1, z_1) = (1, 2, 3)$, $(y_2, z_2, x_2) = (-5, 20, -7)$ gives $|u_y|, |u_z| \sim 10^{-11}$ but $|u_x| \approx 15$ at $t = 15$). In the standard Pecora–Carroll receiver the x-equation is self-driven, $\dot{x}_2 = -\sigma x_2 + \sigma y_2$. Then

$$\dot{u}_x = -\sigma u_x + \sigma u_y,$$

a stable linear equation ($\sigma > 0$) forced by the exponentially decaying input u_y, so

$$u_x(t) = e^{-\sigma t} u_x(0) + \sigma \int_0^t e^{-\sigma(t-s)} u_y(s) \, ds,$$
$$|u_x(t)| \le e^{-\sigma t}|u_x(0)| + \sigma|u(0)| \int_0^t e^{-\sigma(t-s)} e^{-\kappa s} \, ds \;\longrightarrow\; 0.$$

So $u_x \to 0$ as well. With that reading the full receiver (x_2, y_2, z_2) globally synchronizes with the sender (x_1, y_1, z_1), which is the conclusion Step 8 states.



One-way coupled Lorenz systems, $\sigma = 10$, $r = 28$, $b = 8/3$. $|u_y|$ and $|u_z|$ decay exponentially in both cases; $|u_x|$ decays only for the self-driven receiver $\dot{x}_2 = -\sigma x_2 + \sigma y_2$ (right), and levels off at a nonzero value for the literal printed equation (left).

Summary

Step	Result
1–2	A has eigenvalues a (direction $(1,1)$) and $a-2c$ (direction $(1,-1)$); $\lvert x-y\rvert = \lvert x_0-y_0\rvert e^{(a-2c)t}$; sync iff $c > a/2$.
3	M has eigenvalues λ_i and $\lambda_i - 2c$ (eigenvectors $(v,v),(v,-v)$; $M \sim \operatorname{diag}(A, A-2cI)$).
4	$u = v_1 - v_2$ obeys $\dot{u} = (A-2cI)u$.
5	If $\operatorname{Re}\lambda_i < 2c$ for all i, 0 is globally asymptotically stable: $\lvert v_1 - v_2\rvert \to 0$.
6	$J' = AJ$, $K' = (A-2cI)K$, $K(t) = J(t)e^{-2ct}$.
7	Exponents of the difference are $\lambda_i - 2c$; $c > \lambda_1/2 \Rightarrow \lVert K(t)\rVert \le Ce^{-\beta t}$: local synchronization.
8	$E = u_y^2 + u_z^2$ has $\dot{E} = -2u_y^2 - 2bu_z^2 < 0$; $\lvert u(t)\rvert \le \lvert u(0)\rvert e^{-\min\{1,b\}t}$: global one-way synchronization of (y,z).